2.1 Introduction

As a signatory of the Nagoya protocol, Bangladesh government is committed to declare 10 percent of its EEZ as an MPA by the year 2020. When the commitment was made, the EEZ was roughly about 60,000 km^2, but now it has increased to 118,813 km^2 due to the final settlement of maritime border disputes with neighboring states Myanmar and India in 2012 and 2014 respectively (MoFA, 2014). The length of Exclusive Economic Zone (EEZ) has also been increased from 200 nautical mile to 354 nautical miles. The expanded area of EEZ has become important both from ecological, biodiversity and economic viewpoints. The EEZ part of the Bay of Bengal is home to 475 species of fish, 17 species of marine reptiles, 11 species of marine mammals, 20 species of birds, 28 species of crabs (including brackish water and fresh water), 4 species of lobster, 33 species of shrimp, 437 species of marine and brackish water mollusks and 165 species of marine algae and sea weed, which have been identified so far. The mangrove ecosystem adjacent to coastal areas of south-western Bangladesh is very important as breeding and nursing grounds for many marine species. In addition to providing habitats to numerous species, the EEZ of Bangladesh also supports livelihoods for about eleven million people in the coastal zone.

The EEZ is a relatively less studied habitat for biodiversity and there is inadequate administrative and management control over exploitation of renewable biological resources. Currently, natural gas survey and exploration is ongoing under the Energy and Mineral Resources Division (EMRD) of Ministry of Power, Energy and Mineral Resources, which also needs to be operated in a planned way to avoid destruction of biological resources.

2.2 Importance and Benefits of Marine Protected Areas

MPAs, like any other protected areas, are regions in which human activity has been placed under restrictions in the interest of conserving the natural environment, its surrounding waters and the ecosystems, and any cultural or historical resources that may require preservation or management. Marine protected areas' boundaries always include some area of ocean, even if it is only a small fraction of the total area of the territory. Traditionally, marine resources are protected by local, state, territorial, native, regional, or national authorities and this may differ substantially from nation to nation. This variation includes different limitations on development, fishing practices, fishing seasons and catch limits, mooring facilities, bans on removing or disrupting marine life of any kind.

Traditional ferry boats moored at a ghat in Saint Martin's island

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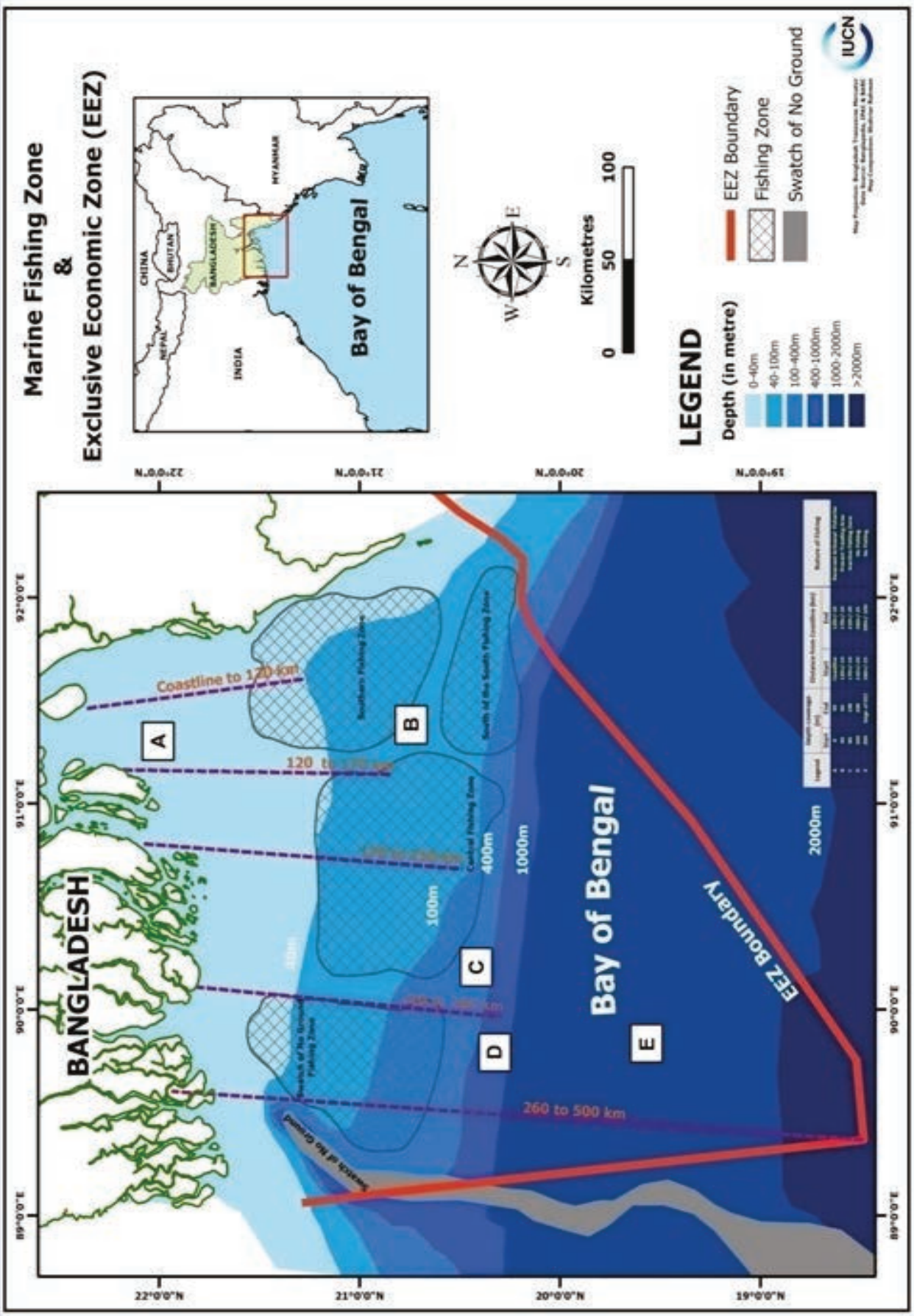

Marine fishing zone and EEZ

Fishers all over the world are proponents of conserving the marine resources upon which they depend as they believe an MPA will enrich their fishing grounds, but they object when it comes to sacrificing profit margins. In Bangladesh fisher groups raise objections whenever a protected area is proposed. It was also observed during national workshops, regional meetings and focus group discussions. This is partly due to the lack of understanding of MPAs and their ultimate goal and partly due to the lack of awareness.

The coastal region is the most unstable, vulnerable and naturally disaster prone area of the country. There are human settlements throughout the coastal region except in the Sundarbans. The coastal areas are now heavily populated and sustain their livelihoods on field crops and resources from the sea. The coastal areas covered with mangroves are declared as reserved forest, and people are not allowed to live there. However, thousands of people are involved in subsistence livelihoods by harvesting/collecting various resources from the Sundarbans Reserved Forest (SRF). The coastal areas resource or mangrove forest resource consumers, particularly the primary users are from the 19 district and 51 upazilas in the coastal region under six greater districts. We need to first identify the natural resource users before declaring/designating MPAs.

We have scanty information on land holding by the vulnerable groups living in coastal zones as newly formed chars are usually occupied by the local elite. A detailed study covering the entire coastal zone has not yet been carried out. Scattered studies by local and international development agencies have shown that for populations living in peripheral areas of Sundarbans the average land holdings of all SRF actors is low (less than one acre or 88 decimals and half an acre or 49 decimals for ownership and operation respectively (IUCN, 2012). The poverty situation of coastal districts shows a dismal picture compared to rest of the country. Islam, 2010 studied head count ratios (HCR) for the Sundarbans adjacent districts and upazilas (SIZ), and showed a much higher poverty rate (0.42) compared to non-SIZ upazilas in Bangladesh (0.26). Although agriculture is still the mainstay of the economy in the region, the SRF provides varied sources of livelihoods to SIZ people which are not commonly available in other parts of the country.

Unfortunately, Bangladeshi people's attitude towards public or community resources are not positive for preservation and restoration. People consider public property as everybody's property when it comes to exploitation, but consider it as nobody's property when comes to preservation or restoration. This attitude in a densely populated country is not conducive for conservation of fragile natural resources and ecosystems, unless strict monitoring and stringent measures are taken against violators.

2.3 Economic Value of Marine Protected Areas

Bangladesh is a land scarce country housing over 160 million people on a landmass of only 14.4 million ha. Natural resource exploitation rate in Bangladesh is one of the highest in the world. If natural resources are not utilized in a sustainable manner then the gradual decline of renewable natural resources can not be arrested. By a recent estimate, roughly 17 percent of the country's land mass seasonally or perennially comes in contact with salt or hypo-saline water and this zone of roughly 2.5 million ha is termed coastal land. Coastal people not only depend on land for their survival, they also harness natural resources from the sea for their livelihoods. Economic information on the sea and its content and importance for the wellbeing of Bangladesh's coastal dwellers is not an easy task to quantify. There is a common assumption in Bangladesh that the BOB is a mine of unlimited source of resources. This is ofcourse not true; the only unlimited resource in the Bay of Bengal contains is perhaps salt. Besides salt, all other living or non-living resources are finite and thus will end one day, if they are not exploited in a sustainable manner.

Prospect of maximum sustainable yield (MSY), from EEZ under no-intervention plays an important role in formulating management plans for protection of ecosystem, conservation strategies for biodiversity, habitat and particular species. Besides, promoting programs for the conservation of biodiversity and ecosystem services is also important.

Currently the size of EEZ constitutes almost 70 percent of the size of Bangladesh but fish landing constitutes the same 23 percent of total production.

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Hilsa is an iconic fish of the Bengali culture

There has been no organized evaluation of the probable exploitable quantity of fish. The total fish production (in MT) from the marine waters is shown in Table 1. This amount of harvested biomass is roughly 9.1 ton per square kilometer of the EEZ,

Table 1. Total landing from EEZ of Bangladesh (2010-2011); Source DOF

Species/Group	Volume in ton
Hilsa	225,325
Other marine fish/shrimp	321,000
Total	**546,333**

calculated on the old area. If the expanded sea area in EEZ is taken into consideration, unit area production is only 4.9 ton/km^2. We know that the territorial waters are overexploited and that the rest of the EEZ is underexploited. The expanded EEZ may be explored and deployed to harvest the unexploited marine resources to increase overall landings. Research and surveys by DoF and BFRI are not adequate and lack resources.

The marine products can be broadly divided into six categories: fish, shrimp, crabs, shells, sea weeds and salts. DOF in its routine publications usually does not mention salt as it is not considered a product of fisheries. A recent publication by DOF (DOF, 2011) indicated that roughly 11.1 million people obtain all or part of their income from marine resources. Several million fishermen living in coastal areas have fishing as their only livelihood option. The marine fishermen are usually landless. With the introduction of shrimp farming in the coastal area, a big number of the landless tenant farmers and their families are now engaged in seasonal harvesting of shrimp, prawn and mullet fingerlings to be used as seeds in shrimp polyculture.

2.4 History and Current Status of Marine Protected Areas in Bangladesh

There is very little information available about the status of protected areas in the marine environment in Bangladesh. While some of the country's terrestrial protected areas encompass parts of the coastal zone, there are no explicit 'marine protected